Organisational Emergence Cosmology

DEBA Theory

Determined by Phase Actualisation

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1 Introduction: Necessity and Contingency of Emergence

The substrate is an absolute void traversed by fluctuations whose nature is intrinsically probabilistic — prior to any physics, any symmetry, any geometry. No rule pre-exists. The sole property of the substrate is that every non-zero actualisation potential is actualised.

This principle follows from the absolute atemporality of the substrate. Time is a quantity internal to bubble-universes; it does not exist in the substrate. In the absence of time, no filter maintains configurations in the state of unrealised possibilities. Everything IS simultaneously.

The DEBA theory articulates three distinct levels:

1. **Necessity** — that bubble-universes exist. Emergence is inevitable in the strict sense: an atemporal substrate cannot contain a threshold without that threshold being crossed, since everything IS. This is the meaning of *Determinics*.

2. **Contingency** — what each bubble-universe is. The phase transition is radically random: no universe is predetermined, each is simply a possible configuration that has been actualised.

3. **Determination** — how each bubble-universe is structured. Once the transition is complete, constants, metric and physical laws are entirely fixed by the state of the actualisation field Φ at that instant.

2 Pre-metric Substrate

DEBA theory proposes that physical reality emerges from a fundamentally atemporal organisational substrate $(\mathcal{C}, \mu)$, where:

- $\mathcal{C}$ is the set of all possible organisational configurations;

- μ is a fluctuation amplitude function encoding the intensity with which each configuration manifests in the substrate — and thereby the condition of what will or will not be actualised.

In this substrate, no geometric structure is postulated: no metric, no space, no time, no physical causality. Relations between configurations are purely organisational: what will be or will not be.

3 Fundamental Organisational Dynamics

The dynamics of the actualisation field Φ, a field variable over the configuration space, is modelled by a functional Langevin equation:

$$d\Phi = -\frac{\delta V}{\delta \Phi}\, ds + \sigma\, dW \tag{1}$$

where $V(\Phi)$ is the organisational potential, σ the fluctuation intensity, and W a Wiener-type process.

The parameter s is an **organisational index**, not to be identified with physical time. It indexes the degree of structuration reached by a given configuration of the field Φ in the space $\mathcal{C}$ — all its values coexist simultaneously in the substrate.

4 Organisational Flash: Condensation and Rupture

Definition 1 (Organisational Flash)**.** *An organisational flash is a phase transition occurring when global coherence $R[\Phi]$ and organisational mass $M[\Phi]$ simultaneously exceed their critical thresholds. The flash instant is formalised as a stochastic stopping time:*

$$\tau_{\text{flash}} = \inf\big\{\, s \geq 0 \;\big|\; R[\Phi(\cdot, s)] \geq R_{\text{crit}} \;\; \textit{and} \;\; M[\Phi(\cdot, s)] \geq M_{\text{crit}} \big\}. \tag{2}$$

At τ_{flash}: rupture. Φ^* condenses. Before: no metric, no time, no physics. After: metric, internal time, physical laws — the Friedmann equations become definable from this instant only.

The flash is not a gradual transition — it is a rupture. Isotropy is not a principle of the substrate: the bubble emerges with no imposed symmetry whatsoever.

Note on proper constants.[1]

5 Bubble-Universe and Emergence of the Metric

After the flash, the condensed configuration Φ^* defines a bubble-universe B. The spatial curvature k is not postulated — it is **extrinsic**, determined by the immersion geometry of the bubble in $\mathcal{C}$. The bubble-universe is of codimension 2, embedded in a substrate without metric.

In the condensed phase, the internal coherence field $s(x)$ becomes definable within the bubble, enabling the emergence of an effective metric:

$$g_{\mu\nu}(x) = f\big(s(x),\, \nabla s(x)\big), \tag{3}$$

where f encodes geometric emergence as a manifestation of coherence gradients.

Apparent Big Bang, without singularity. In DEBA, the Big Bang corresponds to the entry into the geometric phase after the flash:

$$t = 0 \iff s = \tau_{\text{flash}}. \tag{4}$$

Prior to this threshold, geometry is not a defined object.

6 DEBA Cosmological Equations

6.1 Justification of the Friedmann Approach

The Friedmann equations are not postulated in DEBA — they are **recovered in the condensed phase** as an effective description of the bubble B, under the assumption of

[1] The flash corresponds to a coherence beam; each universe arising from it carries a distinct portion of coherence. The proper and singular constants of each universe depend on this carried portion. The probability that two universes share identical constants is vanishingly small.

large-scale homogeneity and isotropy. They are valid only after τ_{flash} and constitute an approximation of the complete organisational dynamics. In the limit $\Phi \to \Phi_*$, $G_{\text{eff}} \to G$, $Q \to 0$, standard cosmology is recovered as a degenerate particular case of DEBA.

6.2 Effective Homogeneity and Isotropy Assumption

At large scales, the bubble B admits an effective FLRW description characterised by a scale factor $a(t)$, a spatial curvature k (extrinsic, cf. Section 5), and an expansion rate:

$$H(t) = \frac{\dot{a}(t)}{a(t)}. \tag{5}$$

6.3 DEBA $\leftrightarrow$ Cosmological Content Dictionary

The effective energy is decomposed into three contributions:

- ρ_b: standard baryonic matter;
- ρ_{coh}: stabilising coherence density (effective dark matter);
- ρ_{adapt}: global adaptive density (effective dark energy).

These last two are the two manifestations of a single coherence field $\Phi(t)$: its spatial projection (gradients) and its temporal adaptive response to expansion.

6.4 Effective Gravitation

$$G_{\text{eff}}(t) = G\,\Gamma\big(\Phi(t)\big), \qquad \Gamma(\Phi) = 1 + \xi\,(\Phi - \Phi_*), \tag{6}$$

where Φ_* is the coherence attractor and ξ a coupling parameter, with $\Gamma(\Phi_*) = 1$.

6.5 DEBA Friedmann Equations

Friedmann-1:

$$H^2 = \frac{8\pi G_{\text{eff}}}{3}\big(\rho_b + \rho_{\text{coh}} + \rho_r\big) + \frac{\Lambda_{\text{eff}}}{3} - \frac{k}{a^2}, \tag{7}$$

with $\Lambda_{\text{eff}}(\Phi) \equiv 8\pi\, G_{\text{eff}}(\Phi)\, \rho_{\text{adapt}}(\Phi)$.

Friedmann-2:

$$\frac{\ddot{a}}{a} = -\frac{4\pi G_{\text{eff}}}{3}\Big[\big(\rho_b + \rho_{\text{coh}} + \rho_r\big) + \frac{3\,(p_b + p_{\text{coh}} + p_r)}{c^2}\Big] + \frac{\Lambda_{\text{eff}}}{3}. \tag{8}$$

In the DEBA interpretation: $p_{\text{coh}} \approx 0$ (effective dark matter behaviour) and $p_{\text{adapt}} \approx -\rho_{\text{adapt}}\, c^2$ (effective dark energy behaviour).

6.6 Conservation Equations with Organisational Coupling

$$\begin{aligned}
\dot{\rho}_b + 3H\left(\rho_b + \frac{p_b}{c^2}\right) &= 0, \\
\dot{\rho}_{\text{coh}} + 3H\left(\rho_{\text{coh}} + \frac{p_{\text{coh}}}{c^2}\right) &= +Q, \\
\dot{\rho}_{\text{adapt}} + 3H\left(\rho_{\text{adapt}} + \frac{p_{\text{adapt}}}{c^2}\right) &= -Q,
\end{aligned} \tag{9}$$

where the organisational exchange term is:

$$Q = \alpha\,\dot{\Phi}\,H. \tag{10}$$

Dark matter and dark energy can thus be seen as two regimes of a single organisational content, redistributed according to the global coherence state $\Phi(t)$.

6.7 Global Coherence Dynamics

$$\dot{\Phi} = -\gamma\,(\Phi - \Phi_*) + \eta\,H + \xi(t), \tag{11}$$

where Φ_* is the coherence attractor, $\gamma > 0$ an organisational relaxation rate, η a sensitivity to expansion, and $\xi(t)$ a residual fluctuation term (negligible at cosmological scales).

6.8 Constitutive Closures

$$\rho_{\text{coh}}(\Phi) = \rho_{\text{coh},0}\left(\frac{\Phi}{\Phi_0}\right)^m a^{-3}, \qquad \rho_{\text{adapt}}(\Phi) = \rho_{\Lambda,0}\left(\frac{\Phi}{\Phi_0}\right)^n. \tag{12}$$

The indices m and n are **free parameters** of the theory, not derivable a priori, to be constrained by observations. In the limit $m = n = 0$, standard ΛCDM cosmology is recovered as a degenerate particular case.

6.9 Boundary Condition: Dissolution of the Bubble

The bubble B exists as long as the adaptive equilibrium condition is satisfied. When expansion dissipates coherence beyond what the organisational flux can compensate:

Rupture. The bubble undergoes a final phase transition. Metric, time, physical laws cease. Nothing physical remains.

7 Dark Matter and Dark Energy: Two Aspects of a Single Field

Within the DEBA framework, dark matter and dark energy are not two independent substances — they are the two manifestations of a single organisational coherence field $s(x,t)$. Dark matter is its spatial projection (gradients); dark energy is its temporal projection (adaptive response to expansion).

This unification naturally dissolves the **cosmic coincidence problem**: the comparable magnitudes of dark matter and dark energy densities today is no longer a coincidence requiring explanation, but the direct consequence of their shared organisational origin.

8 Physical Finiteness and Pluri-Matrix Cosmology

In pluri-matrix cosmology, infinity has no physical existence. Each bubble-universe U_i is a finite local condensation:

$$U_i = \{\, x \in \mathcal{C} \mid s(x) \geq s_{\text{crit}} \,\}. \tag{13}$$

No global inter-bubble metric exists: bubbles are not separated by physical space — their relations take the form of stochastic couplings in regions of shared coherence.

9 Antimatter as Non-Actualised Potentiality

DEBA proposes a radically different answer to the baryon asymmetry problem: **there was never a symmetry to break.** Matter and antimatter are two structurally distinct categories from the substrate itself.

Definition 2 (Antiparticle)**.** *An antiparticle is a configuration $\bar{x} \in \mathcal{C}$ such that:*

$$\forall\, s \geq 0 : \quad s(\bar{x}, s) < s_{\text{crit}}, \tag{14}$$

where $s(\bar{x}, s)$ is the coherence of the associated potentiality region. The coherence of the antiparticle never crosses the threshold required for physical emergence.

This sub-threshold condition is not a transient state — it is a structural property inscribed in the substrate via the coherence kernel $K(\bar{x}, y)$ and the amplitude function $\mu(\bar{x})$. Baryon asymmetry is **structurally organisational**: inscribed in the structure of $K(x, y)$ and μ from the origin. No primordial symmetry exists, hence nothing to break.

10 Observational Signatures

10.1 Cold Spot

The Cold Spot corresponds to limited coherence sharing between bubble-universe U_1 and an adjacent bubble U_2 during the early phases of actualisation:

$$B_1 \cap B_2 \neq \varnothing \implies s(x, \tau) < s_{\text{crit}}, \tag{15}$$

resulting in a local deficit of organisational density in the cosmic microwave background.

10.2 Axis of Evil

The Axis of Evil reflects a residual orientation arising from the organisational coupling with other bubbles:

$$K(x, y) \;\to\; K(x, y) + \epsilon\, \hat{n} \cdot (x - y)\, K_0(x, y), \tag{16}$$

where $\hat{n}$ describes the preferred coupling direction and ϵ the amplitude of the effect.

10.3 Primordial Black Holes

Primordial black holes are **organisational fossils** of inter-bubble overlap — not standard density fluctuations, but coherence transfers at bubble boundaries prior to recombination. Their spatial distribution reflects the organisational geometry of the flash: non-random, correlated with large-scale structures, potentially anisotropic along the inter-bubble coupling direction.

10.4 Status of These Signatures

These signatures constitute coherent explanations within the DEBA framework, falsifiable by observation. They are not exclusive predictions, but convergent markers whose joint identification would constitute a consistent body of evidence compatible with the organisational dynamics.

11 Summary

The DEBA theory describes a pluri-matrix cosmos where:

1. The pre-metric substrate $(\mathcal{C}, \mu)$ — an absolute atemporal void with intrinsic fluctuations — simultaneously contains all configurations. Everything IS.
2. The actualisation field $\Phi(x, s)$ structures the space of organisations under the combined effect of entanglement and resonance.
3. The flash τ_{flash} is a rupture: before, no physics; after, metric, internal time and effective constants emerge entirely from Φ^*.
4. Dark matter and dark energy are the two aspects of a single coherence field: spatial (gradients) and temporal (adaptive response).
5. Adaptive equilibrium formally constrains the field parameters via $Q = \alpha\dot{\Phi}H$ and dissolves the cosmic coincidence problem.
6. The dissolution of the bubble is the symmetric rupture of the flash: brutal, irreversible, without progressivity.
7. The curvature k is extrinsic — determined by the immersion geometry of the codimension-2 bubble in $\mathcal{C}$.
8. Primordial black holes, the Cold Spot and the Axis of Evil are fossil signatures of the organisational geometry of the flash, falsifiable by observation.
9. Baryon asymmetry is structurally inscribed in $K(x, y)$ and μ from the origin — no CP symmetry breaking is required.

A Constructive Questions and Internal Responses

$K(x, y)$ and proper constants. Each bubble-universe arising from the same flash carries a distinct portion of coherence. The probability that two universes share identical constants is vanishingly small.

Adaptive equilibrium and rapid expansion. When coherence flux can no longer compensate for the expansion, the bubble undergoes a final phase transition: rupture, dissolution, nothing physical remains.

Antimatter and instanton trajectories. No stable crossing of the threshold s_{crit} by an antiparticle configuration is possible. Matter–antimatter annihilation is a rare instanton trajectory, reproducible only under extreme artificial conditions.

Overlap signatures and primordial black holes. Primordial black holes constitute fossils of coherence transfer at inter-bubble boundaries. Their non-random spatial distribution constitutes a distinct and falsifiable observational signature.